



Análisis descriptivo

Aprendizaje Automático 1

Librerías:

```
import pandas as pd
import numpy as np
from scipy.stats import zscore
import matplotlib as plt
import seaborn as sns
```

Dataset:

	Unnamed: 0	fake	profile_pic	ratio_numlen_username	len_fullname	ratio_numlen_fullname	sim_name_username	len_desc	extern_url	private	num_posts	num_followers	num_following
0	0	0	Yes	0.27	0	0.0	No match	53	No	No	32	1000	955
1	1	0	Yes	0.00	2	0.0	Partial match	44	No	No	286	2740	533
2	2	0	Yes	0.10	2	0.0	Partial match	0	No	Yes	13	159	98
3	3	0	Yes	0.00	1	0.0	Partial match	82	No	No	679	414	651
4	4	0	Yes	0.00	2	0.0	No match	0	No	Yes	6	151	126

Valores nulos:

```
fake          0
profile_pic   0
ratio_numlen_username  0
len_fullname  0
ratio_numlen_fullname  0
sim_name_username  0
len_desc      0
extern_url    0
private       0
num_posts     0
num_followers 0
num_following 0
```

```
nan_count = df.isna().sum()
```

Estadísticas descriptivas:

```
print(df.describe(include='all'))
```

#Se puede acceder por partes:

```
print(df['num_followers'].count())
```

```
print(df['num_followers'].mean())
```

```
print(df['num_followers'].std())
```

```
print(df['num_followers'].min())
```

```
print(df['num_followers'].quantile(  
q=0.25))
```

```
>>> print(df.describe(include='all'))
```

	Unnamed: 0	fake	profile_pic	ratio_numlen_username	len_fullname	ratio_numlen_fullname	sim_name_username	len_desc	extern_url	private	num_posts	num_followers	num_following
0	0	0	Yes	0.27	0	0.0	No match	53	No	No	32	1000	955
1	1	0	Yes	0.00	2	0.0	Partial match	44	No	No	286	2740	533
2	2	0	Yes	0.10	2	0.0	Partial match	0	No	Yes	13	159	98
3	3	0	Yes	0.00	1	0.0	Partial match	82	No	No	679	414	651
4	4	0	Yes	0.00	2	0.0	No match	0	No	Yes	6	151	126

```
>>> print(df.describe(include='all'))
```

	Unnamed: 0	fake	profile_pic	ratio_numlen_username	len_fullname	ratio_numlen_fullname	sim_name_username	len_desc	extern_url	private	num_posts	num_followers	num_following
count	576.000000	576.000000	576	576.000000	576.000000	576.000000	576	576.000000	576	576	576.000000	5.760000e+02	576.000000
unique	NaN	NaN	2	NaN	NaN	NaN	3	NaN	2	2	NaN	NaN	NaN
top	NaN	NaN	Yes	NaN	NaN	NaN	No match	NaN	No	No	NaN	NaN	NaN
freq	NaN	NaN	404	NaN	NaN	NaN	303	NaN	509	356	NaN	NaN	NaN
mean	287.500000	0.500000	NaN	0.163837	1.460069	0.036094	NaN	22.623264	NaN	NaN	107.489583	8.530724e+04	508.381944
std	166.421153	0.500435	NaN	0.214096	1.052601	0.125121	NaN	37.702987	NaN	NaN	402.034431	9.101485e+05	917.981239
min	0.000000	0.000000	NaN	0.000000	0.000000	0.000000	NaN	0.000000	NaN	NaN	0.000000e+00	0.000000e+00	0.000000
25%	143.750000	0.000000	NaN	0.000000	1.000000	0.000000	NaN	0.000000	NaN	NaN	0.000000	3.900000e+01	57.500000
50%	287.500000	0.500000	NaN	0.000000	1.000000	0.000000	NaN	0.000000	NaN	NaN	9.000000	1.505000e+02	229.500000
75%	431.250000	1.000000	NaN	0.310000	2.000000	0.000000	NaN	34.000000	NaN	NaN	81.500000	7.160000e+02	589.500000
max	575.000000	1.000000	NaN	0.920000	12.000000	1.000000	NaN	150.000000	NaN	NaN	7389.000000	1.533854e+07	7500.000000

Outliers:

```
q_low = df["num_followers"].quantile(0.01)
```

```
q_hi  = df["num_followers"].quantile(0.99)
```

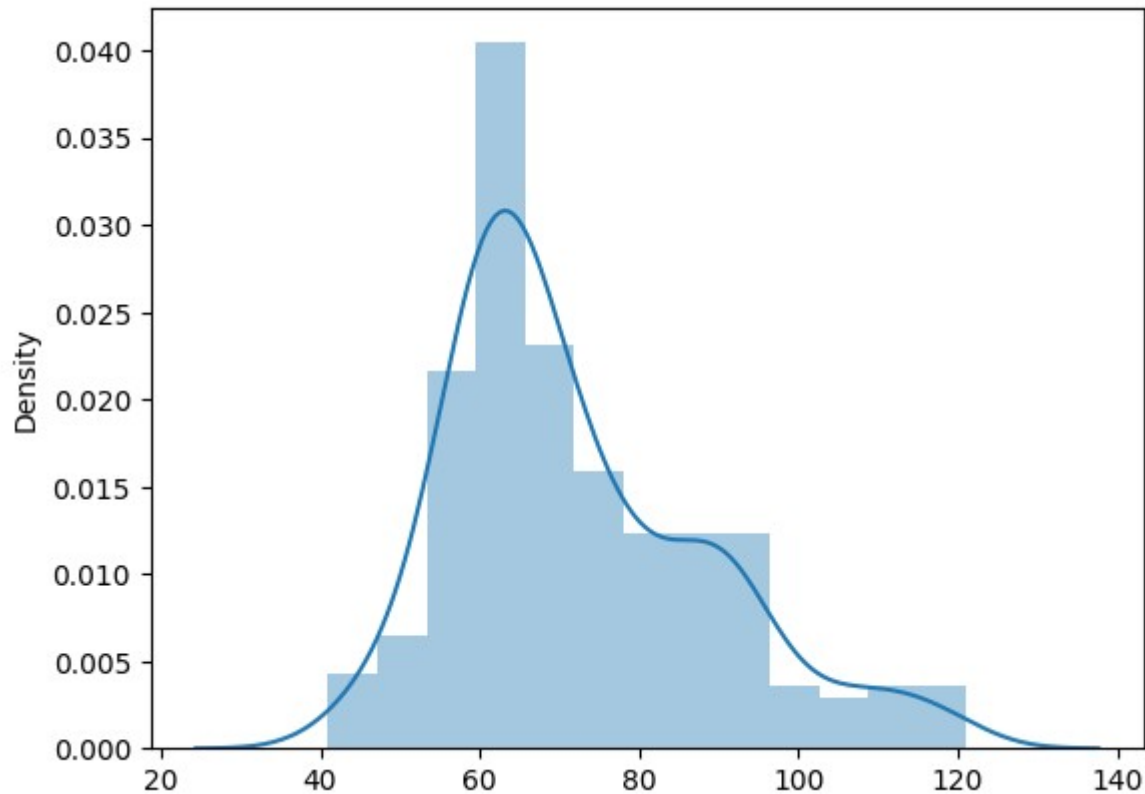
```
df_filtered = df[(df["num_followers"] < q_hi) & (df["num_followers"] > q_low)]
```

```
print(len(df. index))-> 576
```

```
print(len(df_filtered. index)) -> 552
```

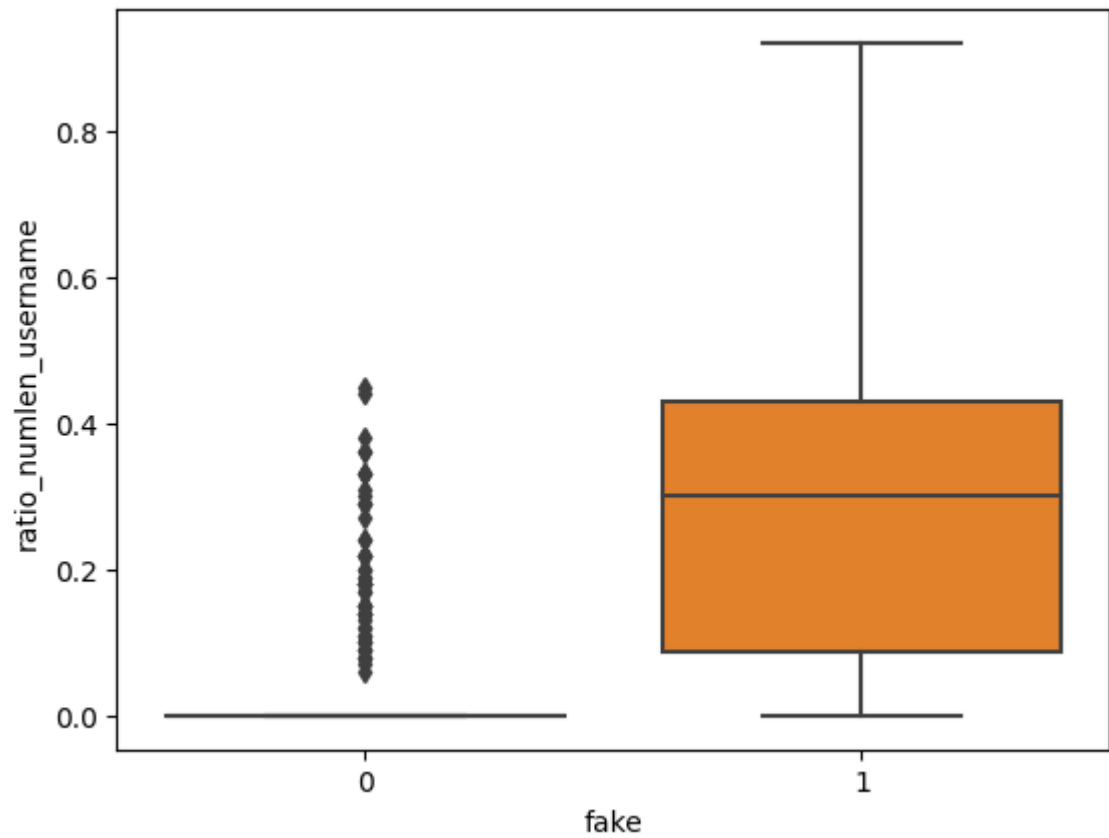
Gráficos:

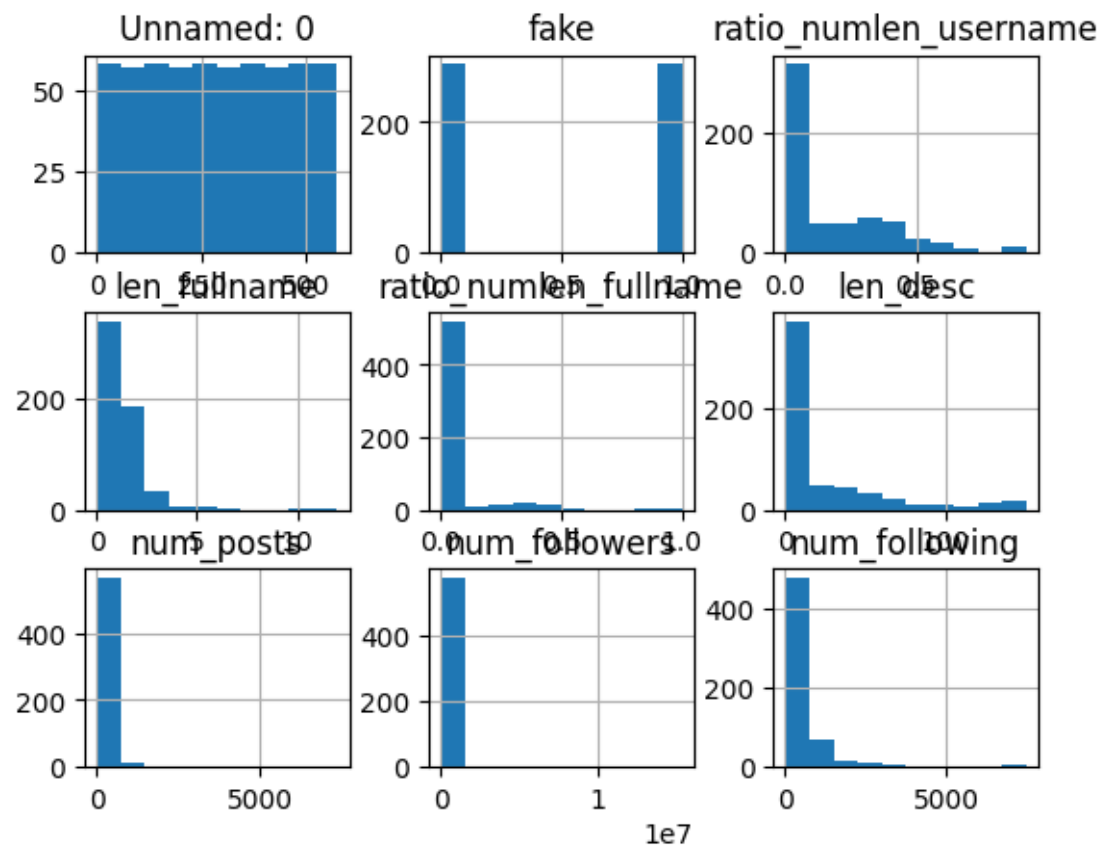
```
sns.distplot(df["ratio_numlen_usern  
ame"], kde=True, rug=False)
```



```
sns.boxplot(x='fake',  
y='ratio_numlen_username', data=df)
```

A





```
df.hist()
```

```

df_numeric = df[['num_followers',
'num_following',
"ratio_numlen_username", "len_fullname",
"ratio_numlen_fullname", "len_desc",
"num_posts"]]

corrmatrix = df_numeric.corr()

print(corrmatrix)

sns.heatmap(corrmatrix, vmax=1, annot=True,
linewidths=.5)

```

